

## Sai Lakshmi Aarthi Vadlapatla

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### EDUCATION

#### UNIVERSITY OF CENTRAL OKLAHOMA

Edmond, OK

MS in Computer and Science- Full Stack Development (Current GPA – 3.2)

Aug 2023- May 2025

Course Work: Incident Analysis & Resp, Internet of Things, Software Engineering II, Practicum In Computing Science, Front End Web Programming, Enterprise Web Programming, Cloud Web Apps Development, Mobile Apps Programming, Software Engineering I and Applications Database Systems

#### VELLORE INSTITUTE OF TECHNOLOGY

Amaravati, India

Bachelor of Technology in Computer Science and Engineering (CGPA – 7.78)

Sep 2019 - May 2023

Course Work: Data Structures and Algorithms, Object Oriented Programming, Design and Analysis of Algorithms, Applied Statistics, Data Analysis, Operating Systems, Machine Learning, Computer Networks, Network Programming, Mobile Application Development, Web Technologies, IOT, Advanced Data Analytics, Introduction to Cryptography, Cloud Computing, Database Management.

### SKILLS

**PROGRAMMING LANGUAGES:** Java, Python, HTML, CSS, SQL, JavaScript, TypeScript.

**SOFTWARE DEVELOPMENT:** OOP, Agile(Scrum), JIRA, REST API

**DATABASES & DATA WAREHOUSING:** PostgreSQL, SQLite, MySQL.

**DATA VISUALIZATION:** Power BI, Tableau.

**DATA ENGINEERING & ANALYTICS:** Data Modeling, Data Warehousing, ETL, Pandas, NumPy.

**FRAMEWORKS:** Django, Spring Boot Flask, Angular, React.

**TECHNOLOGIES:** AWS, Docker, Linux

### WORK EXPERIENCE

#### QUIDDITY SOFTWARE SOLUTIONS PVT. LTD.

Hyderabad, India

DATA ANALYST/ SOFTWARE ENGINEER INTERN

Jan 2023 - Jul 2023

- Analyzed and validated client datasets with 98% accuracy using **SQL** for **data cleansing, transformation, and mapping**.
- Developed **Tableau** dashboards to track data integration **KPIs** and uncover trends in client implementation success
- Identified and resolved 95% of data discrepancies by collaborating with engineering and customer success teams
- Documented business rules and **data validation** processes to streamline future client onboarding
- Contributed to app configuration testing, increasing pre-launch data reliability by 30%
- Gained experience with **RESTful APIs, SFTP, and tools like JIRA, Confluence, and FME**

#### QUIDDITY SOFTWARE SOLUTIONS PVT. LTD.

Hyderabad, India

SOFTWARE ENGINEER INTERN

Oct 2022- Dec 2022

- Developed a **RESTful service-oriented architecture (SOA)** using **Spring Boot**, optimizing search efficiency for new and used goods across platforms like Craigslist and eBay.
- Built a **secure authentication system** with **Spring Security and JWT**, ensuring role-based access control and data protection.
- Designed and implemented a **scalable backend** with **Spring Data and MySQL Enterprise Edition**, enhancing data retrieval and system performance.
- Integrated **encryption techniques** and **AWS services** to enhance security and scalability.
- Developed a **responsive web and mobile UI** using **Angular, Ionic Framework, and TypeScript**, improving user experience.

### UNIVERSITY PROJECTS

#### SYLLABUS CHATBOT USING RETRIEVAL AUGMENTED GENERATION(RAG)

Dec 2024

- Developed a **web-based Syllabus Chatbot** using **Retrieval Augmented Generation (RAG)**, enabling **natural language interaction** for course syllabi search.
- Implemented **Flutter & Dart** for the frontend, integrated **Firebase (Firestore, Storage, Authentication)** for secure syllabus management, and utilized **syncfusion\_flutter\_pdf** for **text extraction**, supporting **Gemini/OpenAI APIs** for intelligent syllabus retrieval.

#### TREE CLASSIFICATION USING AERIAL IMAGES

Oct 2022

- Applied **deep learning** models such as **VGG Net, ResNet, Xception, SVM, and YOLOv5** to analyze aerial images for identifying endangered trees in India, supporting deforestation prevention.
- The project involved comparative analysis of multiple machine learning models for effective tree classification.

#### HEART DISEASE PREDICTION USING MACHINE LEARNING

Sep 2022

- Developed a **machine learning-based diagnostic system** in **Python** to predict heart disease risk using a dataset with 13 medical features, including age, blood pressure, and cholesterol levels.
- The model leverages algorithms to assess the likelihood of heart disease based on clinical data.

#### WATER MONITORING FOR GARDENING

July 2022

- Developed an automated water management system for plant and crop irrigation using **Arduino Uno** and a **soil moisture sensor**, ensuring optimal water supply.
- The system, built with **Java**, continuously monitors soil moisture content and adjusts irrigation based on real-time data.

#### FINGERPRINT VOTING SYSTEM

Nov 2022

- Designed and implemented a **fingerprint biometric-based voting system** using **Java** and **MySQL**, automating the election process while preventing fraudulent activities.
- This system enhances security and accuracy by authenticating voters through unique fingerprint data.